1. Concept Difference

Restore Database

- Used when you have a **backup file (.bak)**
- Creates a database from backup
- Used for:
    - Disaster recovery
    - Moving DB between servers
    - Point-in-time recovery

Detach / Attach Database

- Used when you have **actual database files (.mdf, .ldf)**
- Detach removes DB from server (files stay)
- Attach brings DB back using files
- Used for:
    - Moving DB quickly between servers
    - No backup required

---

2. Practical Scenario 1 – Restore Database

Situation:

You have a backup file:

C:\Backup\SchoolDB.bak

Step-by-Step

1. Restore database

```
RESTORE DATABASE SchoolDB
FROM DISK = 'C:\Backup\SchoolDB.bak'
WITH REPLACE,
MOVE 'SchoolDB_Data' TO 'C:\Data\SchoolDB.mdf',
MOVE 'SchoolDB_Log' TO 'C:\Data\SchoolDB.ldf';
```

Explanation:

- FROM DISK → backup file location
- MOVE → where new files will be stored
- REPLACE → overwrite if DB already exists

Practice Question (Restore)

Q1: Restore database CompanyDB from

D:\Backup\CompanyDB.bak

into:

- MDF → D:\SQLData\CompanyDB.mdf
- LDF → D:\SQLData\CompanyDB.ldf

Q2: What happens if you restore a DB without WITH MOVE on a different server?

---

3. Practical Scenario 2 – Detach & Attach

Situation:

You want to move a database to another server using files.

Step 1: Detach Database

```
USE master;
GO
EXEC sp_detach_db 'SchoolDB';
```

This removes DB from SQL Server

Files remain in folder:

- .mdf
- .ldf

Step 2: Copy Files

Copy:

C:\Data\SchoolDB.mdf

C:\Data\SchoolDB.ldf

to new server

Step 3: Attach Database

```
CREATE DATABASE SchoolDB
ON
(FILENAME = 'C:\Data\SchoolDB.mdf'),
(FILENAME = 'C:\Data\SchoolDB.ldf')
FOR ATTACH;
```

Practice Question (Detach/Attach)

Q3: Write query to attach DB SalesDB using:

- E:\DB\SalesDB.mdf
- E:\DB\SalesDB_log.ldf

Q4: What will happen if .ldf file is missing during attach?

---

4. Real-Life Comparison

Feature	Restore	Detach/Attach
Requires Backup	Yes (.bak)	No
Uses MDF/LDF files	No	Yes
Safe for recovery	Highly safe	Risky
Use case	Disaster recovery	Quick migration

Supports point-in-time	Yes	No

---

5. When to Use What

Use Restore when:

- You want **safe recovery**
- You need **backup strategy**
- Moving between environments securely

Use Detach/Attach when:

- You have **no backup**
- Need **fast transfer**
- Working in **test/dev environments**

---

6. Bonus Scenario (Interview Style)

Question:

Your production database crashed. You have:

- .bak file
- .mdf file

Which method should you use?

Answer: Use **Restore** (backup is safest and consistent)

---

7. Hands-On Mini Lab (Try This)

Step 1: Create DB

CREATE DATABASE TestDB;

Step 2: Backup

BACKUP DATABASE TestDB
TO DISK = 'C:\Backup\TestDB.bak';

Step 3: Detach

EXEC sp_detach_db 'TestDB';

Step 4: Attach Again

CREATE DATABASE TestDB
ON (FILENAME = 'C:\Program Files\Microsoft SQL Server\MSSQL\Data\TestDB.mdf'),
  (FILENAME = 'C:\Program Files\Microsoft SQL Server\MSSQL\Data\TestDB_log.ldf')
FOR ATTACH;

Step 5: Restore (Rename DB)

RESTORE DATABASE TestDB_Restore
FROM DISK = 'C:\Backup\TestDB.bak'

```sql
WITH MOVE 'TestDB' TO 'C:\Data\TestDB_Restore.mdf',
   MOVE 'TestDB_log' TO 'C:\Data\TestDB_Restore.ldf';
```

Quick Summary

- **Restore = Backup-based recovery (safe & professional)**
- **Detach/Attach = File-based movement (fast but risky)**